

Subscription Autopilot

An AI agent that runs your recurring service spending on Arc — and can prove it behaved.

Ignyte × Circle × Arc · Track 4 (Agentic Economy) · Ronie Joseph ·
roniejosephv@gmail.com · Arc Testnet, chain 5042002, gas = USDC

Circle's official sample shows an agent that *can* pay. Autopilot is an agent that pays **well** — and every dollar it moves passes through an out-of-process policy authority. **The agent can want anything; it can only pay what policy allows.**

WHAT IT DOES

RESEARCH

Probes competing x402 sellers' live price menus (free quotes) against real usage history.

NEGOTIATE

Compares volume/loyalty tiers across sellers at every renewal — switched at 21.9% cheaper per call in the demo.

EXECUTE

Gasless USDC nanopayments per call (x402), batch-settled by Circle Gateway on Arc.

THE CUSTODY BOUNDARY

UNTRUSTED — THE AGENT

LLM, injectable. No keys, no Circle SDK, no chain RPC — only POST /pay with a stated reason. A compromised agent can ask; it cannot *take*.

CUSTODY — SPENDGUARD

allowlist → perTx → monthly → perService
→ velocity → threshold

allow → pay · hold → human · deny → code

Policy versions and spend epochs are auto-anchored on Arc every 10 minutes (SpendAnchor.sol), so every decision is externally verifiable — a tamper-evident audit of the agent's behavior.

CIRCLE STACK — REAL, LOAD-BEARING INTEGRATIONS

Nanopayments / x402	Core rail — per-request dynamic pricing; policy enforced in the onBeforePaymentCreation hook, inside SpendGuard's trusted process.
Circle Wallets (DCW)	Key custody — EIP-712 signing on a developer-controlled EOA; no raw private key exists anywhere in the stack — payments <i>and</i> on-chain anchoring alike.
Circle Gateway	Settlement + treasury — gas-free EIP-3009 authorizations, batch-settled onchain, transfer UUID receipts.
USDC on Arc	Settlement asset <i>and</i> gas token — audit-anchor transactions are themselves paid in USDC.
CCTP · USYC · StableFX	Autonomous cross-chain treasury (CCTP, built) + tokenized-yield and FX legs (USYC / StableFX) wired behind flags, testable on access.

VERIFIED ON ARC TESTNET — LIVE DEPLOYMENT

✓ Full E2E loop (Day-0 gate, verify-day0.sh) ✓ Keyless payments + on-chain anchoring verified (DCW-signed) ✓ All four demo beats live in production — metering, 21.9% switch, injection denied, human hold/release ✓ SpendAnchor live: 0xf550a882da3c26fbacd1b68aa83867102206b143

autopilotdashboard-production.up.railway.app (live demo) ·
github.com/roniejosephv-star/subscription-autopilot · Apache-2.0 · testnet only